

```
git pull
./repo sync -d
```

The `-d` option switches the various projects' `HEAD` back to their Android manifest revision (i.e., the default mainstream repository version). This is helpful if you made modifications to the repos (and have associated source changes) but want to revert back to the master version temporarily. It will leave your staged or working directory changes untouched.

Building the B2G tree

To start the build process, run the following command in the terminal inside the B2G folder:

```
./build.sh
```

It will take around one hour or more to build the B2Ge source code locally. Once you are done building the source code, you can start the emulator by running the following command in the terminal in the B2G folder itself:

```
./run-emulator.sh
```

Submitting bug reports on B2G, Firefox OS and Gaia

Once you've got a B2G/Firefox OS build running, you'll probably want to start to file bug reports against this specific

version, so that the Firefox OS community can improve things as effectively as possible. You'll need to file the bug on Bugzilla, under the 'Firefox OS' project, but you should include specifics about the version:

- To start with, give the major version number, e.g., 1.4.0.0-pre-release. This can be found on the device under *Settings > Device Information*
- You can provide more specific version identifiers by returning the current hashes from the Gaia and Gecko repositories. This can be done as follows:

```
#!/bin/bash
(cd gaia; echo "gaia $(git rev-parse HEAD)")
(cd gecko; echo "gecko $(git rev-parse HEAD)")
```

After making some changes in the B2G tree, you need to push them to the master tree, which is based on Git. **END** 

References

[1] https://developer.mozilla.org/en-US/Firefox_OS/

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Given a set of questions based on world events, and access to the World Wide Web (and its search engines), can you come up with a program that generates answers to the questions based on search query results? One possible way is to do a keyword-based search on the search query results and return the document passage that bears the most similarity. What are the other possible techniques you can use? For instance, consider the following two questions:

- What is the major food item exported by India?
- What is the only city in India which, apart from being the capital of a state, is also located on the banks of a river and is on the seashore?

How do these two questions vary in terms of difficulty in finding answers through an automated system? How would your solution deal with question (b)?

By the way, I wanted to point our readers to an interesting article I read recently in the *IEEE Spectrum* magazine by Prof Jordan (of machine learning, EM algorithm and Bayesian networks fame) titled 'Delusions of Big Data and Other Huge Engineering Efforts' which is available at:

<http://spectrum.ieee.org/robotics/artificial-intelligence/machinelearning-maestro-michael-jordan->

on-the-delusions-of-big-data-and-other-huge-engineering-efforts. He makes the important point that the inferences drawn from Big Data need to be validated for whether they are random patterns found by analysis (causality does not imply causation) or real root-causes explaining the data patterns. On being asked what research area he would target if he got a billion dollar research grant, he picked Natural Language Processing.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, Happy New Year and happy programming! **END** 

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The author is an expert in systems software and is currently working with Hewlett Packard India Ltd. Her interests include compilers, multi-core and storage systems. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group Computer Science Interview Training India at <http://www.linkedin.com/groups?home=&gid=2339182> &HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"gid=2339182